Maple Thermodynamics Academy

Unit 2: States of Matter and Phase Transitions

Student textbook | All competition levels | Batch 2

Seven complete chapters, 17 worked numerical examples, original particle/phase/graph figures, formula packet, glossary and unit review.

Read the chapters in order. Use the separate WS07–WS13 practice packet after each corresponding chapter. The unchanged Unit 1 materials remain the prerequisite reference.

All plotted H, Y and X data are hypothetical educational models. No laboratory heating or pressurized equipment is needed.

Contents

1. States of matter and particle behavior

2. Phase transitions

3. Heating and cooling curves

4. Sensible heat

5. Latent heat and multistage changes

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This is a complete Unit 2 volume within a curriculum still in production.

U2-C01 - States of matter and particle behavior

Unit 2 | All competition levels | public-student | Rule alignment: 5.c.ii, printed B57

Learning objectives

Describe solids, liquids and gases with a particle model; connect structure to shape, volume and compressibility; distinguish a model drawing from actual particle size.

Prerequisites

Unit 1 heat, temperature, systems and density; division and volume units.

Start with an observation

An ice cube keeps its shape on a plate. Liquid water spreads across the plate. Water vapor spreads through the available air. These are three physical states of the same substance. The molecules are still H_2O: a state change is not a chemical decomposition into hydrogen and oxygen. What changes is how the molecules are arranged, how freely they can move, and how important their interactions are relative to their motion.

A particle model replaces an enormous number of atoms or molecules with a few circles and motion arrows. It helps us explain patterns we cannot see directly. The circles are symbols; they are not photographs or accurate scale drawings. A dot does not become a larger molecule when the material is heated. The spaces and the movements may change instead. When comparing drawings of the same sealed sample, preserve particle number unless matter is explicitly added or removed.

Solids: motion without flowing freely

In a solid, particles remain near relatively fixed positions while vibrating. Many familiar solids are crystalline: their average positions form a repeating pattern. Not all solids have a perfect repeating lattice; glass is a useful example of a noncrystalline solid. A solid generally resists a change of shape and volume because moving its particles far from their usual arrangement requires energy. Saying the particles are fixed in place is acceptable only if you also remember their vibrations.

Heating a solid usually increases the energy of its microscopic motion and changes its average spacing somewhat. At the melting transition, added energy can instead help change the arrangement into a liquid. This is why a particle model must include interactions as well as motion. Do not predict the temperature from particle separation alone.

Liquids: close neighbors that rearrange

Liquid particles remain close to neighboring particles, but their arrangements change continuously. A liquid flows and takes the shape of the part of its container that it occupies. It has an approximately fixed volume under modest pressure changes. Flowing therefore does not imply widely separated particles: a liquid flows even though its particles are crowded together.

Both liquids and solids are usually difficult to compress substantially because much of the space is already occupied by closely packed particles. Their volumes are not mathematically constant under every condition; they can expand, compress, or change structure. “Nearly incompressible” is a useful model for many introductory questions, not an absolute law.

Gases: widely separated particles and collisions

Gas particles are far apart on average compared with their sizes. Between collisions they move freely through the available space. Their collisions with container walls produce pressure. A gas has neither a fixed shape nor a fixed volume independent of its container. If the container allows more space, the gas spreads into it; if a piston reduces the available volume, the gas can become much denser without making each molecule smaller.

At the same temperature, particles of different gases need not have the same typical speed because their masses differ. For the same dilute gas, a higher absolute temperature corresponds to a higher average translational kinetic energy. That connection supports the ideal-gas model in Chapter U2-C07. It does not make temperature a direct measure of total internal energy.

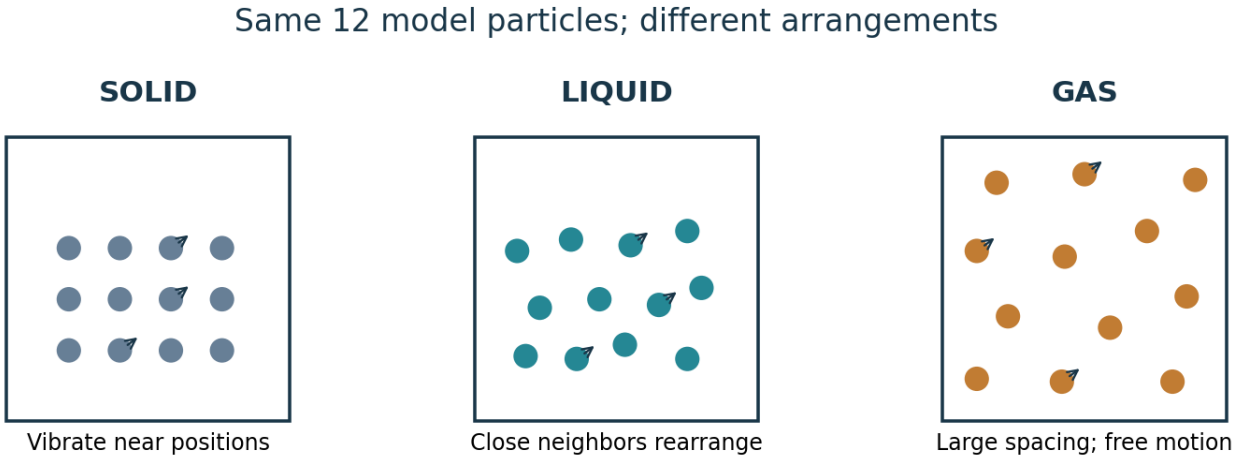
Connect the model to measurements

Use mass and volume to calculate density. When a sealed sample changes state, its total mass stays the same even if its volume changes. Gases commonly occupy much more volume than the same mass of liquid at ordinary conditions, giving them much smaller densities. For many substances a solid is denser than its liquid, but ordinary ice is an important exception: its structure occupies more volume per mass than liquid water. Never use “solid is always densest” as a rule.

Particle drawings can support a claim but cannot replace measured conditions. If two pictures show different dot counts and different box sizes, you need both pieces of information to compare number per volume. If a gas-filled box doubles in volume with the same number of molecules, the average number per volume halves. The molecules have not vanished; they are more spread out.

Plasma is another state of matter involving substantial ionization. This unit focuses on the solid, liquid and gas phase relationships used in the supplied syllabus. No plasma calculations are needed for this chapter.

Visual model



Each panel has 12 model particles. Close packing in the liquid is intentional. Particle sizes are schematic, not to scale.

Definitions and units

Solid: material that resists shape change and usually maintains volume. Liquid: material that flows while maintaining approximately fixed volume at modest pressure changes. Gas: material that expands to fill available volume. Compressibility: how readily volume decreases under pressure. Density ρ: mass per volume, such as g/mL. Particle number N: a count, not a mass.

Equations and their meaning

ρ=m/V, with m in g and V in mL for g/mL. Number per volume=N/V. These ratios describe the whole sample; they do not imply that molecules change size. 1 L=1000 mL.

Worked example: Density after melting

Given: A hypothetical sealed sample has mass 45.0 g, solid volume 50.0 mL and liquid volume 45.0 mL. No matter escapes.

Find: Density in each state.

Equation: ρ=m/V.

Substitution: ρ_s=45.0 g/50.0 mL; ρ_l=45.0 g/45.0 mL.

Arithmetic: 45.0/50.0=0.900; 45.0/45.0=1.00.

Answer with units: Solid: 0.900 g/mL; liquid: 1.00 g/mL.

Physical interpretation: This illustrative sample is denser as a liquid. Its mass is unchanged; volume changed. The numbers are not a material identification.

Worked example: Count density in a gas model

Given: A diagram uses 24 dots for a sealed gas in 3.0 L. It expands to 6.0 L with no particles escaping.

Find: Dots per liter before and after.

Equation: Model number density=N/V.

Substitution: Before: 24/3.0; after: 24/6.0.

Arithmetic: 24/3.0=8.0; 24/6.0=4.0.

Answer with units: 8.0 model dots/L initially; 4.0 model dots/L finally.

Physical interpretation: The representation becomes half as crowded. A dot represents many real molecules, so these are diagram counts, not an actual molecular concentration.

Common confusions

Solid particles move; liquid particles are not necessarily far apart; gas molecules do not grow to fill a larger container. A sealed phase change conserves mass. Density rankings need conditions and material identity.

Real-life connection

A sealed flexible package can change volume when its contents change temperature or state. The package shape alone does not tell you whether matter was created. In the competition cup, the water is mainly a liquid whose changing temperature is measured; gases around it behave differently.

Recap

Shape, volume and compressibility arise from particle arrangement, movement and interactions. Keep particle identity and mass distinct from spacing and sample volume.

Review questions

1. Why can a liquid flow while remaining hard to compress?

2. What produces pressure on the walls of a gas container?

3. Why is “solid particles do not move” incorrect?

Practice

Open WS07. Complete the chapter examples first, then attempt the worksheet independently.

Mastery checklist

☐ I can draw three states without changing particle identity.

☐ I can explain gas compression using spacing.

☐ I can calculate and compare densities without assuming solids are always densest.

U2-C02 - Phase transitions

Unit 2 | All competition levels | public-student | Rule alignment: 5.c.ii, printed B57

Learning objectives

Name and distinguish six phase transitions; identify energy absorbed or released; explain evaporation, boiling and condensation using particles.

Prerequisites

U2-C01; Unit 1 heat-flow direction, systems and mass bookkeeping.

A phase change keeps the substance recognizable

Melting ice produces liquid water, not a new chemical substance. Freezing reverses that change. A physical phase transition rearranges matter between phases without changing the chemical identity of the substance. The surrounding conditions matter: a statement such as “water boils at 100 °C” needs the approximate pressure of 1 atmosphere and a pure-water context.

The six named transitions connect solid, liquid and gas. Melting is solid to liquid; freezing is liquid to solid. Vaporization is liquid to gas; condensation is gas to liquid. Sublimation is solid directly to gas, and deposition is gas directly to solid. Vaporization includes both evaporation and boiling. The arrow direction determines the name. A student should be able to name both directions on every connection, rather than memorize a list without a particle explanation.

Why some transitions absorb energy

To melt a solid, energy must be supplied to change the ordered or constrained particle arrangement into a liquid arrangement. To vaporize a liquid, energy must separate particles much farther against their attractions. Sublimation also requires an energy input under the usual conditions considered here. We call melting, vaporization and sublimation endothermic phase changes for the substance: they absorb heat from surroundings when driven thermally.

Freezing, condensation and deposition release energy to the surroundings under the reverse conditions. They are exothermic for the substance. During condensation, particles become more closely associated and energy leaves. During freezing, a more constrained solid arrangement forms and energy is released. Freezing does not mean that cold is a substance entering the sample.

Remember whose energy transfer you describe. If water vapor condenses on a cooler surface, the condensing water releases energy and the surface can gain energy. Those opposite statements are consistent because their system boundaries differ. Signs in later equations will follow this boundary choice: heat entering the selected sample is positive.

Evaporation is not the same as boiling

Evaporation happens at a liquid's surface and can occur below the boiling temperature. Some surface molecules have enough energy to leave the liquid. If higher-energy molecules escape preferentially and their energy is not immediately replaced, the remaining liquid can cool. Air movement and humidity influence the net evaporation rate, so a single temperature is not enough to predict drying time.

Boiling involves vapor bubbles forming and growing within the liquid. It occurs when the liquid's vapor pressure reaches the surrounding pressure, with a small depth-related pressure effect in real containers. Bubbles in boiling pure water contain water vapor, not separated hydrogen and oxygen gases. At lower external pressure, water can boil at a lower temperature. This matters for cooking at elevation; it is not a change in the identity of water.

Visible white mist near a kettle is mostly tiny liquid droplets formed after vapor cools and condenses. Water vapor itself is invisible. Fogging a mirror is condensation, not liquid water leaking out of the glass. Frost can form by deposition of water vapor directly onto a sufficiently cold surface; not every frost crystal began as a visible liquid drop.

Coexistence and reversibility

A sample can contain more than one phase at once. Ice and liquid water can coexist at the melting temperature under suitable pressure. At that temperature, adding energy can increase the liquid fraction rather than increase temperature. Removing energy can increase the solid fraction. The next chapters make these energy amounts quantitative.

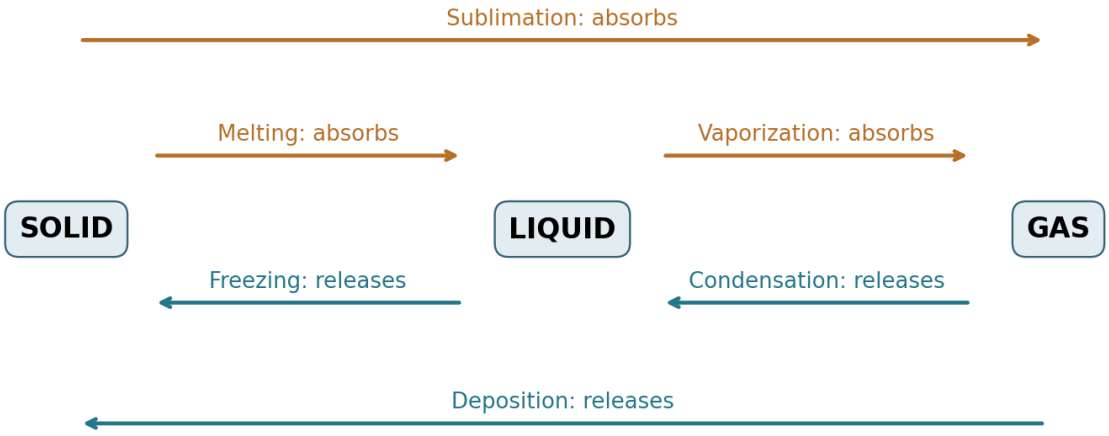
At equilibrium, forward and reverse molecular transfers can occur at equal average rates. In a closed container, vaporization and condensation may balance without the molecules stopping. Outside equilibrium, one direction dominates. Avoid assuming that seeing a few droplets proves the entire sample has reached a uniform equilibrium state.

Track matter as well as energy

An open dish may lose liquid mass by evaporation, but the water has not been destroyed: it has entered the surrounding air. For a sealed system including all phases, the total mass stays constant. Fractions of the sample in different phases add to the total. Mass percentages and balances are useful before learning latent heat: they tell you how much material has completed the transition and how much remains.

All examples here are paper models. Handling boiling water, hot vapor, pressure vessels or dry ice is not required for this curriculum. Recorded classroom observations and diagrams provide the necessary evidence without a home high-temperature experiment.

Visual model



Upper energy-absorption arrows and their lower energy-release reverses are distinguished by labels; color is supplementary.

Definitions and units

Phase: a physically distinct, relatively uniform form of matter. Endothermic: absorbs energy as heat into the selected sample. Exothermic: releases energy as heat. Vaporization: liquid to gas. Sublimation: solid to gas without an intervening liquid phase. Deposition: gas to solid. Vapor pressure: equilibrium pressure of a substance’s vapor above its condensed phase at a given temperature.

Equations and their meaning

m_total=m_solid+m_liquid+m_vapor when those are the included phases. Fraction changed=m_changed/m_total; percent changed=100×fraction. Do not confuse a mass fraction with a temperature reading.

Worked example: Partial melting

Given: A sealed 80.0 g water sample initially consists entirely of ice. After partial melting, 30.0 g is liquid.

Find: Ice remaining and percentage melted.

Equation: m_ice=m_total−m_liquid; percent melted=100m_liquid/m_total.

Substitution: m_ice=80.0−30.0 g; percent=100×30.0/80.0.

Arithmetic: 80.0−30.0=50.0; 100×0.375=37.5.

Answer with units: 50.0 g ice remains; 37.5% of the mass has melted.

Physical interpretation: Both phases are still present. No mass disappeared as the arrangement changed.

Worked example: Net phase exchange

Given: Over an interval in a closed model, 6.0 g evaporates from liquid and 2.0 g of vapor condenses back.

Find: Net change in liquid mass.

Equation: $\Delta m_{liquid} = m_{condensed} - m_{evaporated}$.

Substitution: $\Delta m_{liquid} = 2.0\ g - 6.0\ g$.

Arithmetic: $2.0 - 6.0 = -4.0$.

Answer with units: Liquid mass decreases by 4.0 g; vapor mass increases by 4.0 g.

Physical interpretation: Both directions can occur simultaneously, while the net change favors vaporization. The total closed-system mass remains constant.

Common confusions

Evaporation does not require boiling. Condensation and freezing release rather than absorb latent energy. Ice melting does not split water molecules. Water vapor is not the visible cloud of liquid droplets.

Real-life connection

Sweat can cool skin as water evaporates. Condensation on a cold cup transfers energy from the condensing vapor to the cup and surroundings. These phase-change routes are separate from merely warming or cooling a fixed liquid mass.

Recap

Learn the six arrow directions, then follow energy and matter across a named system boundary. Coexisting phases and simultaneous opposite transfers do not violate conservation.

Review questions

1. Name the reverse of sublimation and state its usual heat-transfer direction.

2. Why can a wet cloth dry without boiling?

3. What happens to total mass when water evaporates inside a sealed container?

Practice

Open WS08. Complete the chapter examples first, then attempt the worksheet independently.

Mastery checklist

☐ I can name all six transitions from arrows.

☐ I can distinguish evaporation from boiling.

☐ I can track energy direction and phase masses separately.

U2-C03 - Heating and cooling curves

Unit 2 | All competition levels | public-student | Rule alignment: 5.c.ii, printed B57

Learning objectives

Read axes, slopes and plateaus; distinguish temperature change from phase change; calculate interval energy from stated net power without calculus.

Prerequisites

U2-C01–02; subtraction, ratios, minutes to seconds and E=Pt.

What does the graph measure?

A heating curve shows how temperature changes as a sample receives energy. The horizontal axis might be elapsed time or energy supplied. Read it before doing any calculation. A point on a time-temperature graph gives a temperature at a time; it does not directly give joules. If a constant net power into the sample is specified, time can be converted to transferred energy using $Q=P_{net}\Delta t$.

The word net means the transfer that actually reaches and remains available to heat or transform the sample after relevant losses are accounted for. A heater's electrical rating is not automatically this net power. A graph drawn against energy supplied also needs a clear definition of whose supplied energy is on the axis.

The original figure in this chapter is an idealized pure substance H heated at constant pressure, with equilibrium melting at 20 °C and boiling at 80 °C. It is not water and not measured Maple device data. Its temperatures are chosen to make the ideas visible. The sample is assumed uniform, pressure fixed, net input power constant, and superheating or supercooling negligible.

Read the five segments in order

From 0 to 2 minutes, the solid warms from −10 °C to 20 °C. Its temperature rises, so the segment slopes upward. From 2 to 6 minutes, the temperature remains at 20 °C while melting occurs. In the interior of this interval, solid and liquid coexist and the liquid fraction grows. At the endpoints, the phase just begins or just finishes changing.

From 6 to 12 minutes, the entirely liquid sample warms from 20 °C to 80 °C. From 12 to 20 minutes, the temperature holds at 80 °C as vaporization occurs. Liquid and vapor coexist during this plateau. From 20 to 22 minutes, the gas warms from 80 °C to 110 °C. A higher final temperature alone does not tell you how much of the total input was needed for the two transitions.

A plateau is not evidence of zero energy transfer. During an ideal equilibrium transition, energy changes the particle arrangement and phase fraction rather than the temperature. If you wrote $Q=0$ only because $\Delta T=0$, you used the wrong physical model for that segment. Chapter U2-C05 supplies $Q=mL$ for phase-change amounts.

Compare slopes fairly

The average temperature-change rate over an interval is $(T_2-T_1)/(t_2-t_1)$. This is ordinary rise divided by run, not calculus. Use the units of both axes. On the liquid segment, a 60 °C rise takes 6 minutes, so the rate is 10 °C/min. A cooling segment has a negative rate if time increases to the right.

A steeper temperature-time slope can mean smaller heat capacity, but only when the net heating power and other relevant conditions are comparable. A larger heater can make the same sample warm faster. A gas or solid segment with a different slope may reflect different heat capacities, but do not infer material properties from slope alone if power or losses change.

A longer plateau under the same net power means more energy for that transition. Comparing plateau lengths for different masses requires care: a large sample can take longer to melt even when its latent heat per kilogram is unchanged. Time is not itself a material property.

Cooling reverses the energy transfer

A cooling curve can include a downward gas segment, a condensation plateau, a downward liquid segment, a freezing plateau and a downward solid segment. During a plateau, the sample releases energy even though its temperature is constant. At a given fixed pressure, the equilibrium freezing/melting temperature is the same, as is the condensation/boiling temperature.

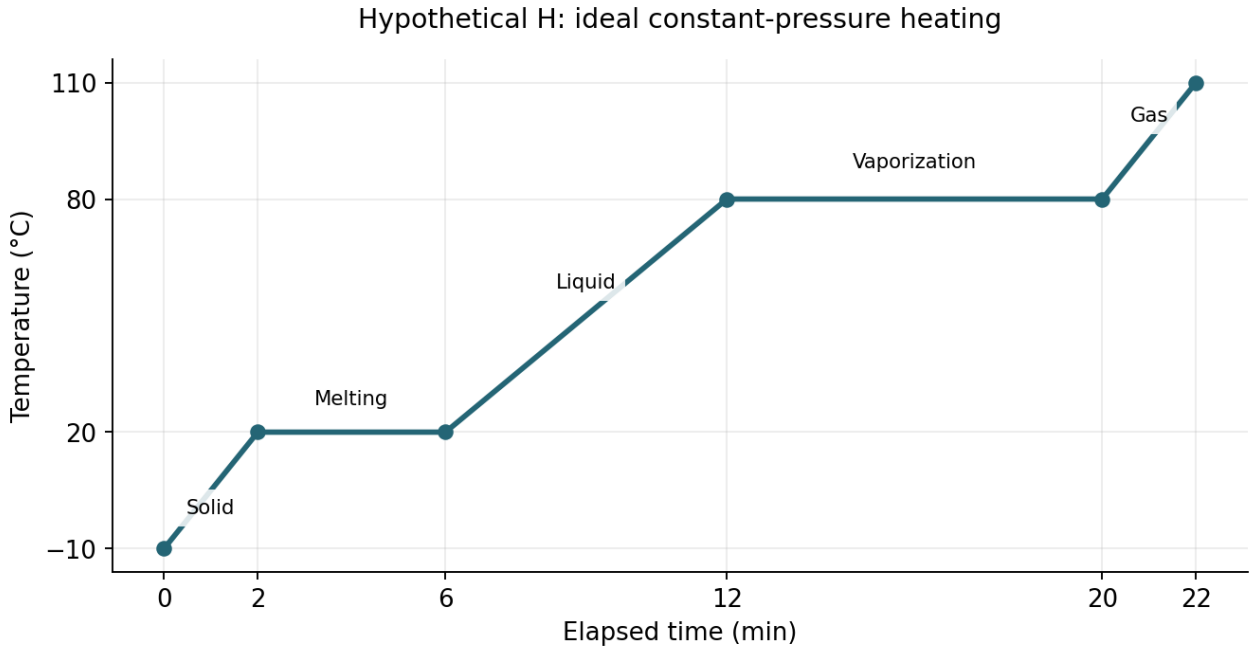
Real recorded cooling data need not show perfect horizontal lines. A liquid may supercool temporarily before crystals form; mixtures can change phase over a temperature range; a probe may lag or sample an uneven temperature. These possibilities explain why the textbook graph is an idealization. They are not reasons to ignore axis labels or assume every flat-looking interval proves a phase change.

Use a graph as evidence, not decoration

Describe a segment using three linked statements: what the graph does, what phase or phases are present under the stated model, and what happens to energy. For example: “At 20 °C from minute 2 to minute 6 the graph is flat; solid and liquid coexist; the input energy melts more solid.” This is stronger than naming only “melting.”

When drawing, label both axes and their units, plot the supplied points accurately, and connect the points only as instructed. If a line segment is declared straight, linear interpolation is allowed. Otherwise the position between measurements is an estimate, not an exact fact. Always identify simulated data as simulated.

Visual model



Data: (0 min, −10 °C), (2, 20), (6, 20), (12, 80), (20, 80), (22, 110). Straight segments are part of the idealized model.

Definitions and units

Heating/cooling curve: temperature versus time or transferred energy. Plateau: an approximately horizontal segment. Average slope: change in vertical quantity divided by change in horizontal quantity. Net power P_{net}: effective heat-transfer rate into the sample, in W. Phase fraction: mass in a selected phase divided by total sample mass.

Equations and their meaning

Average rate$=(T_2-T_1)/(t_2-t_1)$, in °C/min if those are the axes. For constant net power, $Q=P_{net}\Delta t$ with Δt in seconds for J. A temperature plateau requires a phase-change model rather than $Q=mc\Delta T$ alone.

Worked example: Liquid-segment slope

Given: The hypothetical H curve has T=20 °C at 6 min and T=80 °C at 12 min.

Find: Average temperature-change rate.

Equation: Rate=ΔT/Δt.

Substitution: Rate=(80−20) °C/(12−6) min.

Arithmetic: 60/6=10.

Answer with units: 10 °C/min.

Physical interpretation: This is a warming rate, not a heat input in watts. It applies to the stated liquid segment.

Worked example: Energy during a melting plateau

Given: The H melting plateau runs from 2 to 6 min; constant net power into the sample is 100 W.

Find: Heat received during melting.

Equation: $Q=P_{net}\Delta t$.

Substitution: Δt=(6−2)×60=240 s; Q=100 J/s×240 s.

Arithmetic: 100×240=24000.

Answer with units: 24,000 J, or 24 kJ.

Physical interpretation: The sample gains energy even though its temperature stays 20 °C throughout the modeled transition.

Common confusions

A flat temperature line does not mean the heater stopped. Slope units are not energy units. A plateau cannot automatically be assigned to boiling without knowing the material, pressure and surrounding segments. A graph against time is not a pressure-temperature phase diagram.

Real-life connection

In a competition water-heating trial near room temperature, a gradually decreasing slope may result from increasing losses rather than a phase change. Do not label a device heating curve “boiling” merely because it flattens somewhat.

Recap

Read axes first. Separate sloping single-phase segments from ideal phase-change plateaus, and use net power only when its meaning and value are stated.

Review questions

1. What phases coexist inside a melting plateau?

2. Why can a cooling plateau have nonzero heat transfer?

3. What assumption is needed before a longer time interval proves a larger heat input?

Practice

Open WS09. Complete the chapter examples first, then attempt the worksheet independently.

Mastery checklist

☐ I can label all five ideal heating segments.

☐ I can calculate a slope with correct units.

☐ I can use $Q = P_{net}\Delta t$ and explain a nonzero plateau energy.

U2-C04 - Sensible heat

Unit 2 | All competition levels | public-student | Rule alignment: 5.c.ii, printed B57

Learning objectives

Choose and rearrange Q=mcΔT for a single phase; keep units and signs consistent; separate heat received by water from energy supplied to a whole setup.

Prerequisites

U1-C05–06 conversions, U2-C03 graph segments, elementary one-variable algebra.

Start with two cups

Imagine equal-temperature cups holding different masses of the same liquid. Suppose both receive the same heat while remaining liquid. The smaller mass usually changes temperature more. The same energy is shared among less material. Now imagine equal masses of water and another substance. Their temperature changes can differ because materials require different energy amounts per mass per degree.

Specific heat c expresses this response. A large c means a relatively large heat input is needed to raise a given mass by one degree under stated conditions. Specific heat is not thermal conductivity: c concerns energy storage associated with temperature change, while conductivity concerns transfer through a material. Unit 3 will develop that difference.

“Sensible heat” refers to heat associated with a temperature change rather than a phase change. The name does not mean that feeling an object gives a safe or accurate temperature measurement. In this chapter, the useful model is Q=mcΔT. We use it for a specified single phase with c treated as approximately constant over the interval and no extra work or reactions requiring a different balance.

Build the equation from meaning

Suppose a material needs c joules per gram for a 1 °C rise. A mass m grams then needs m×c joules for that same rise. A rise ΔT degrees multiplies that amount again. This gives Q=mcΔT, not because a formula must be memorized first, but because energy scales with mass and temperature change in the stated model.

Water near ordinary classroom temperatures is often modeled with c=4.18 J/(g·°C), equivalent to 4180 J/(kg·K). These are the same approximate specific heat in different units. If you use grams, choose the first form; if kilograms, the second. The 4.18 value is a rounded material property, not an exact conversion constant. Specific heat can vary with temperature and phase, so always use the value supplied in the problem.

The dimensions provide a check: g×J/(g·°C)×°C leaves J. A Celsius temperature change and a Kelvin temperature change have the same numerical size, so either works when paired with the corresponding c units. Absolute temperatures are different: do not replace ΔT with the final Kelvin reading.

Decide the sign before computing

Define Q as positive for heat entering the selected sample. Define $\Delta T = T_{final} - T_{initial}$. Warming gives a positive ΔT and positive Q in this model. Cooling gives both negative. If a problem asks “how much heat is released,” it often wants the positive magnitude $|Q|$; make clear that the sample's signed heat is negative.

For example, cooling from 40 °C to 25 °C means $\Delta T = -15$ °C. Using +15 without saying it is a magnitude can reverse the physical meaning. A complete explanation states either “$Q = -\ldots$ J for the water” or “the water releases … J,” not an unexplained negative number.

Solve for the requested variable

To solve for temperature change, divide both sides of $Q = mc\Delta T$ by mc: $\Delta T = Q/(mc)$. Then $T_{final} = T_{initial} + \Delta T$. To solve for mass, divide by $c\Delta T$: $m = Q/(c\Delta T)$. To find a specific heat from suitable data, $c = Q/(m\Delta T)$. Rearranging one equation for one variable is within the competition mathematics limits.

Before trusting the result, check that the assumed phase remains possible. If an assumed liquid-water calculation at ordinary pressure predicts a temperature far above boiling without accounting for vaporization, the model has crossed its boundary of validity. Split the process into stages instead of treating a phase transition as an ordinary rise.

Keep the surroundings in view

In a real heating setup, energy can warm the cup, device parts and air as well as the water. Energy can also leave the system during heating. $Q_{water} = mc\Delta T$ describes the heat needed for the stated water change under the model; it does not prove that a lamp supplied only that much energy. A device prediction based on perfect absorption is generally an optimistic model, not a validated prediction.

For several single-phase stages, calculate each stage with the correct phase's c. For a transition, use latent heat as taught next. These calculations remain arithmetic: identify a stage, choose its model, compute, and add signed transfers or positive required amounts consistently.

Definitions and units

Specific heat c: energy per mass per temperature-change unit, J/(g·°C) or J/(kg·K). Sensible heat Q: heat associated with temperature change under stated conditions, J. Temperature change ΔT: final minus initial, °C or K. Heat capacity of the sample $C = mc$, in J/°C or J/K.

Equations and their meaning

$Q = mc\Delta T$; $\Delta T = Q/(mc)$; $T_f = T_i + \Delta T$; $m = Q/(c\Delta T)$; $c = Q/(m\Delta T)$. Use a constant-c single-phase model. Positive Q enters the sample. For water use $c = 4.18$ J/(g·°C) unless the problem supplies another value.

Worked example: Warm a water sample

Given: m=150 g; c=4.18 J/(g·°C); water warms from 18.0 °C to 28.0 °C with no phase change.

Find: Q into water.

Equation: $Q=mc(T_f-T_i)$.

Substitution: $Q=150\ g\times4.18\ J/(g\cdot°C)\times(28.0-18.0)\ °C$.

Arithmetic: $\Delta T=10.0\ °C$; $150\times4.18\times10.0=6270$.

Answer with units: $Q=+6.27\times10^3$ J, or +6.27 kJ.

Physical interpretation: This is the modeled water requirement, not necessarily the lamp’s total output.

Worked example: Cooling and a negative sign

Given: m=80.0 g of a material; c=0.900 J/(g·°C); it cools from 55.0 °C to 25.0 °C without phase change.

Find: Signed heat Q and heat-release magnitude.

Equation: $Q=mc\Delta T$.

Substitution: $Q=80.0\times0.900\times(25.0-55.0)$ J.

Arithmetic: $\Delta T=-30.0\ °C$; $80.0\times0.900\times(-30.0)=-2160$.

Answer with units: $Q=-2.16\times10^3$ J; 2.16 kJ is released.

Physical interpretation: The negative sign refers to energy leaving the selected sample.

Worked example: Predict a final temperature

Given: A 100 g liquid-water sample begins at 21.0 °C and receives 2090 J; c=4.18 J/(g·°C); no phase change or other energy terms are included.

Find: Final water temperature.

Equation: $\Delta T=Q/(mc)$; $T_f=T_i+\Delta T$.

Substitution: $\Delta T=2090/(100\times4.18)\ °C$; $T_f=21.0+\Delta T$.

Arithmetic: $100\times4.18=418$; $2090/418=5.00$; $21.0+5.00=26.0$.

Answer with units: $T_f=26.0\ °C$.

Physical interpretation: The modest rise is consistent with liquid water and the stated energy input.

Common confusions

Use final minus initial, not final temperature alone. Convert g and kg consistently. A high c does not automatically mean slow heat transfer. $Q=mc\Delta T$ is not the entire story when a sample changes phase.

Real-life connection

For the event’s 100 mL water, an approximate mass of 100 g may be useful when a problem explicitly permits water density near 1.00 g/mL. Final temperature still depends on how much energy the water actually receives.

Recap

Sensible-heat calculations connect mass, material response and temperature change. The equation works only with consistent units, a stated sign convention and an appropriate single-phase model.

Review questions

1. Why does twice the mass need twice the heat for the same rise, if c is unchanged?

2. How does a negative Q differ from a negative temperature?

3. What additional calculation is needed after obtaining ΔT if the question asks for T_final?

Practice

Open WS10. Complete the chapter examples first, then attempt the worksheet independently.

Mastery checklist

☐ I can select Q=mcΔT only for a suitable stage.

☐ I can solve for Q, mass or temperature.

☐ I can explain the sign and distinguish sample energy from source energy.

U2-C05 - Latent heat and multistage changes

Unit 2 | All competition levels | public-student | Rule alignment: 5.c.ii, printed B57

Learning objectives

Use Q=mL for phase transitions; determine partial melting; combine sensible and latent stages without skipping boundaries.

Prerequisites

U2-C02–04; mass conversion, energy conservation and one-variable rearrangement.

Energy can change phase instead of temperature

Imagine ice and water coexisting at the melting temperature under a fixed pressure. Add a small amount of heat. Some ice melts, but the common equilibrium temperature can remain the same. The energy has not vanished. It changes the particle arrangement and interaction energy and, depending on the process, can also be associated with volume-change work. Heat during a transition should not automatically be identified with internal-energy change alone.

Latent heat L tells us the heat required per unit mass for a specified phase transition at stated conditions. The term “latent” historically suggests hidden: the energy transfer is not revealed by a temperature increase during the ideal plateau. It is not energy that has become impossible to measure. The melted or vaporized mass is another observable consequence.

For a complete transition of mass m, the heat magnitude is mL. We take tabulated L as positive. Melting, vaporization and sublimation require positive Q into the sample when driven thermally; freezing, condensation and deposition have negative Q for the same amount transformed under reverse conditions. If the question asks for the amount released, report the positive magnitude and state its direction.

Fusion and vaporization are different

The latent heat of fusion L_f concerns melting/freezing. Fusion here means melting, not nuclear fusion. The latent heat of vaporization L_v concerns liquid-vapor conversion. For water near the specified ordinary-pressure transitions, use L_f=334 J/g at 0 °C and L_v=2260 J/g at 100 °C in this packet. These are rounded teaching values, not universal constants valid at all pressures and temperatures.

A gram of boiling water requires much more energy to become vapor than a gram of melting ice requires to become liquid. The number 2260 J/g is about 6.77 times 334 J/g. Do not assume every phase change uses the same L, and do not use ice's specific heat in a melting plateau just because ice is present.

Partial transition: compare the budget first

If 10 g of ice at its melting temperature needs 3340 J to melt completely, an input smaller than 3340 J cannot melt it all under the ideal model. The melted mass is m_melted=Q/L_f. The remaining mass is the initial ice mass minus that melted mass. While both phases remain at equilibrium under the same pressure, the temperature stays at the melting temperature.

If the heat budget is larger than the required melting energy, first subtract the melting requirement. Only the remaining energy can warm the newly melted liquid. This order prevents a common mistake: using all the input energy to calculate a rise even though some had to change phase. If the starting ice is below the melting temperature, it must first be warmed using its own specific heat.

Build a multistage energy ledger

For ice initially below 0 °C that finishes as liquid water above 0 °C at ordinary pressure, there are three stages: warm the ice, melt it, then warm the liquid. Write one line per stage. Use c_ice for the first, L_f for the second and c_water for the third. Add the positive heat requirements if the whole process is heating.

If the final state is vapor above the boiling temperature, add two more stages: vaporize the liquid and warm the vapor. Do not apply a single c across ice, liquid and gas. A question will supply the necessary material values. The long problem is a sequence of simple operations, not a demand for calculus.

For a cooling process, the stage order reverses and Q values are negative for energy leaving the sample. One may instead add release magnitudes, but the chosen convention must be clear. Mixing signed heats and positive magnitudes without labeling them can create an impossible net result.

What the model assumes

The neat plateau applies to an idealized pure substance at fixed pressure near equilibrium. A solution or mixture may melt or boil over a range, and real specimens can have temperature gradients. Latent heat depends on conditions. In a classroom heating model, assume no unaccounted energy absorbed by the container and no loss unless the problem gives those terms. In a real device, those assumptions need evidence.

Water's latent heat explains why phase changes can dominate an energy budget. It also explains why a temperature trace alone cannot reveal every transfer: a constant temperature can accompany a large heat flow. However, the competition heat-collection activity is not an invitation to create steam or add ice or other substances to the supervisor's water. These examples are written-test models, not device-construction instructions.

Definitions and units

Latent heat L: transition heat per mass, J/g or J/kg. L_f: latent heat of fusion. L_v: latent heat of vaporization. Partial melting: only part of the solid becomes liquid. Energy budget: available heat compared with the amounts required by successive stages.

Equations and their meaning

For a transition, $|Q|=mL$. For heating transitions $Q=+mL$; for reverse transitions $Q=-mL$. $m_{changed}=|Q|/L$ when all the specified transfer drives that transition. For successive heating stages $Q_{total}=Q_1+Q_2+\ldots$; subtract completed-stage requirements before using the remaining budget.

Worked example: Melt ice already at 0 °C

Given: 20.0 g ice at 0 °C melts to water at 0 °C near 1 atm; $L_f=334$ J/g.

Find: Required heat.

Equation: $Q=mL_f$.

Substitution: $Q=20.0\ \text{g}\times 334\ \text{J/g}$.

Arithmetic: $20.0\times 334=6680$.

Answer with units: $Q=+6.68$ kJ.

Physical interpretation: Temperature does not change; all the specified heat drives melting.

Worked example: Only part of the ice melts

Given: An ideal 40.0 g ice sample begins at 0 °C and receives 5010 J; $L_f=334$ J/g; no other heat sinks.

Find: Melted mass, remaining ice and final temperature.

Equation: $m_{melted}=Q/L_f$; $m_{remaining}=m_{initial}-m_{melted}$.

Substitution: $m_{melted}=5010/334$ g; $m_{remaining}=40.0-m_{melted}$.

Arithmetic: $5010/334=15.0$; $40.0-15.0=25.0$.

Answer with units: 15.0 g liquid and 25.0 g ice at 0 °C.

Physical interpretation: The full-melting requirement is $40.0\times 334=13{,}360$ J, greater than the input. Two phases remain.

Worked example: Warm ice, melt it, then warm the water

Given: 10.0 g ice at −10.0 °C finishes as liquid at 20.0 °C near 1 atm; $c_{ice}=2.10$ J/(g·°C), $L_f=334$ J/g, $c_{water}=4.18$ J/(g·°C).

Find: Total required heat, ignoring container and losses.

Equation: $Q_{total}=mc_{ice}(0-T_i)+mL_f+mc_{water}(T_f-0)$.

Substitution: $Q_1=10.0\times 2.10\times 10.0$; $Q_2=10.0\times 334$; $Q_3=10.0\times 4.18\times 20.0$, all in J.

Arithmetic: $Q_1=210$ J; $Q_2=3340$ J; $Q_3=836$ J; sum=4386 J.

Answer with units: $Q_{total}\approx 4.39$ kJ.

Physical interpretation: The melting stage dominates. Treating the entire 30 °C span with liquid-water c would miss the transition.

Common confusions

$\Delta T = 0$ does not imply $Q = 0$ during a phase transition. L is not c and its units do not include a temperature denominator. “At the melting temperature” does not mean “already melted.” Use the phase fraction to check the final state.

Real-life connection

Phase-change packs can absorb or release energy with modest temperature variation while a transition is underway. That principle is a thermodynamic connection, not a recommendation to add prohibited materials or energy sources to an event device.

Recap

Choose L for the transition and c for temperature-changing stages. Compare the energy budget with each stage requirement before deciding the final phase and temperature.

Review questions

1. Why does a latent-heat unit lack “per degree”?

2. What must happen before ice at $-5\ ^\circ\mathrm{C}$ can begin melting near 1 atm?

3. If some ice remains after heating at equilibrium near 1 atm, what does that tell you about the final temperature?

Practice

Open WS11. Complete the chapter examples first, then attempt the worksheet independently.

Mastery checklist

☐ I can choose L_f or L_v and assign a sign.

☐ I can determine whether a transition is partial or complete.

☐ I can solve a three-stage heat budget with units on each stage.

U2-C06 – Reading phase diagrams

Unit 2 | All competition levels | public-student | Rule alignment: 5.c.ii, printed B57

Learning objectives

Identify phase regions and coexistence boundaries; follow constant-pressure and constant-temperature paths; distinguish triple and critical points from an ordinary boiling point.

Prerequisites

U2-C01–03; coordinate graphs, axis units, interpolation and temperature differences.

A map of equilibrium states

A phase diagram tells you which phase is stable at a specified temperature and pressure. The horizontal axis is commonly temperature, and the vertical axis pressure. This is not a heating curve: neither axis is time, and no line automatically tells you how fast a sample warms. Read the axes before deciding what a movement across the page means.

Choose a point by locating its temperature horizontally and pressure vertically. If it lies inside a solid, liquid or gas region, that phase is stable in the equilibrium model. A point on a boundary represents conditions at which the adjacent phases can coexist. The diagram does not tell you how much of each phase is present there. To find phase amounts, you need additional information such as mass and an energy balance.

The original diagram here represents hypothetical substance X. Its numerical coordinates and straight-line connections are chosen for teaching and are not measurements for water or any identified chemical. Treat the boundary coordinates as the definition of this model. This avoids accidentally reading exact physical constants from a schematic sketch.

Read the three coexistence boundaries

The solid-gas boundary separates solid from gas and describes sublimation/deposition equilibrium. The solid-liquid boundary describes melting/freezing equilibrium. The liquid-gas boundary describes boiling/condensation equilibrium. Moving across a boundary changes the stable phase under a sufficiently slow equilibrium process. The crossing direction determines the transition name.

All three boundaries meet at the triple point. At this unique temperature and pressure for a pure substance, solid, liquid and gas can coexist in equilibrium. “Triple” refers to three phases, not three different substances or three temperatures. Below the triple-point pressure in this ordinary single-component diagram, no stable liquid region is available. Heating a solid at such a pressure takes it directly into the gas region by sublimation.

The liquid-gas boundary ends at the critical point. Beyond the critical temperature and critical pressure, the fluid is supercritical: it lacks a distinct liquid-gas phase boundary of the usual kind. It is not correct to extend an ordinary boiling line forever or to claim that raising pressure alone above the critical temperature necessarily produces a separate liquid phase. The solid-liquid boundary shown need not end at the critical point; that point concerns the liquid-gas distinction.

Follow a path deliberately

At constant pressure, move horizontally. At constant temperature, move vertically. For substance X at 110 kPa, warming from 10 °C to 90 °C begins in the solid region, crosses the melting boundary at 26 °C, passes through the liquid region and crosses the boiling boundary at 60 °C before ending as gas. The exact 26 °C crossing follows the stated straight solid-liquid line, not guesswork from an unlabeled image.

At a constant temperature of 40 °C, increasing pressure from 40 to 110 kPa crosses the liquid-gas boundary at 70 kPa. The stable state changes from gas to liquid. That process is condensation caused by a change of pressure under the model. The temperature can remain constant even though the state changes.

A point at a boundary is not automatically “in the higher-pressure phase.” Both adjacent phases are allowed at equilibrium. A path along a boundary is different from crossing it. State what the point or path does instead of memorizing “up means liquid,” because an upward path can cross different boundaries at different temperatures.

Real substances need their own diagrams

For many substances, the melting line slopes toward higher temperature as pressure increases, corresponding to a solid denser than its liquid under the relevant conditions. Water near ordinary conditions is an important exception: ordinary ice is less dense than liquid water, and its melting line has a negative slope in the usual pressure-temperature plot. Do not use substance X's positive melting slope as a universal law.

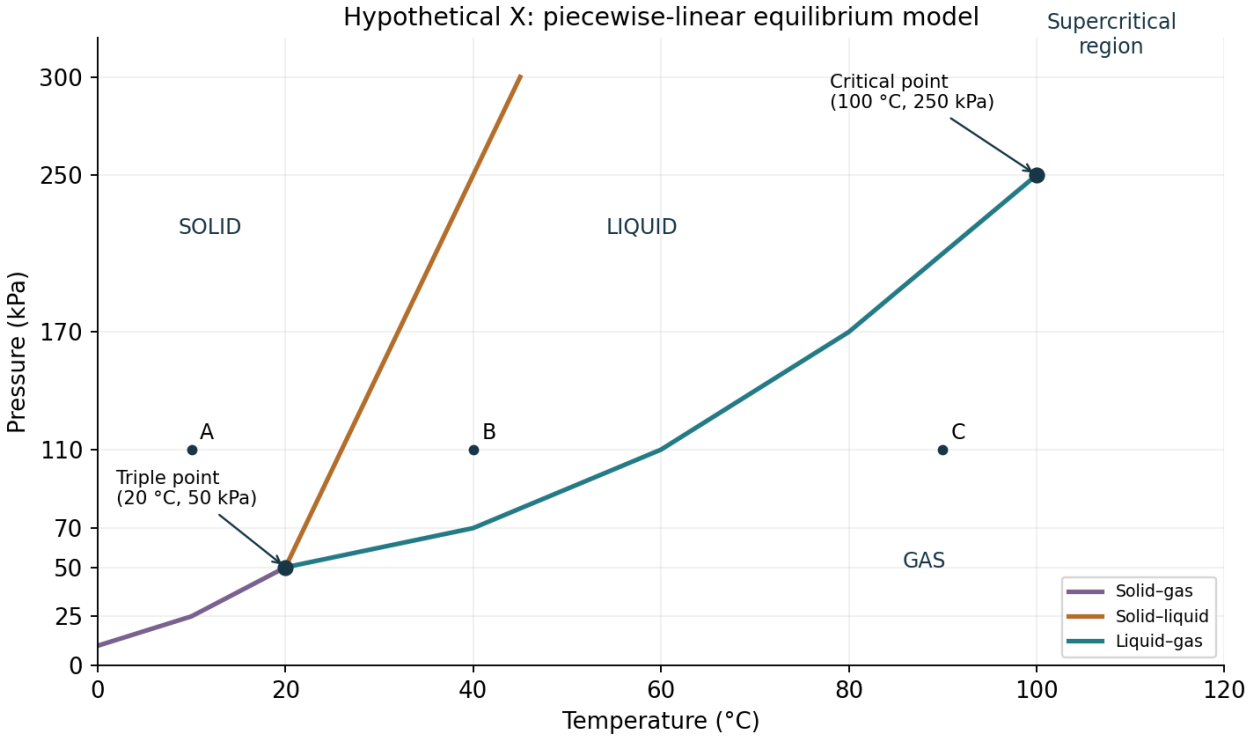
The usual boiling temperature of water near 100 °C refers to approximately 1 atmosphere. On a phase diagram, a normal boiling point is the liquid-gas boundary crossing at 1 atmosphere, not the triple or critical point. Pressure changes the boiling temperature, and real phase diagrams can include additional solid forms at high pressure. This beginner model intentionally focuses on one solid, one liquid and one gas region.

Read numbers without overclaiming

Use supplied values or clearly stated interpolation. The diagram's coordinate list accompanies the figure so an exact problem does not depend on measuring pixels. For a straight segment, proportional interpolation is allowed with elementary ratios. For a real curved boundary, do not assume linearity unless the problem permits an approximation.

A phase diagram supports equilibrium reasoning. It does not specify the heat required, the rate of transition, the amount of substance or the temperature lag of a real experiment. Combine it with other information only after naming the relevant assumptions.

Visual model



Model coordinates: solid-gas boundary (0 °C,10 kPa), (10,25), (20,50); solid-liquid boundary (20,50) to (45,300); liquid-gas boundary (20,50), (40,70), (60,110), (80,170), (100,250). Connect consecutive listed points with straight segments. Triple point=(20 °C,50 kPa); critical point=(100 °C,250 kPa).

Definitions and units

Phase boundary: conditions allowing two adjacent phases to coexist. Triple point: equilibrium coexistence of solid, liquid and gas at a particular T and P. Critical point: endpoint of the liquid-gas coexistence curve. Normal boiling point: boiling temperature at 1 atm. Supercritical fluid: fluid beyond its critical temperature and pressure without a distinct ordinary liquid-gas boundary.

Equations and their meaning

No universal equation generates every phase diagram. For the fictional X model only, the solid-liquid line is $P=10T-150$ with P in kPa and T in °C over the plotted segment $20 \le T \le 45$. Rearrange to $T=(P+150)/10$. This is a teaching-line definition, not a physical law. Other boundary values are given in the diagram table.

Worked example: A constant-pressure liquid range

Given: For X at 110 kPa, the melting line obeys $P=10T-150$ and the table gives boiling at 60 °C.

Find: Melting temperature and width of the stable-liquid temperature interval.

Equation: $T_m=(P+150)/10$; interval width $=T_b-T_m$.

Substitution: $T_m=(110+150)/10$; width $=60-T_m$.

Arithmetic: $260/10=26$; $60-26=34$.

Answer with units: Melting at 26 °C; liquid-region interval width 34 °C (between 26 and 60 °C).

Physical interpretation: At either boundary phases may coexist; strictly inside the interval the stable phase is liquid.

Worked example: Read a vaporization boundary with stated interpolation

Given: The X boundary contains (60 °C,110 kPa) and (80 °C,170 kPa), joined by a straight segment in this model.

Find: Boundary pressure at 70 °C.

Equation: Fraction $=(70-60)/(80-60)$; $P=110+\text{fraction}\times(170-110)$.

Substitution: Fraction $=10/20$; $P=110+(10/20)\times60$ kPa.

Arithmetic: Fraction $=0.5$; $P=110+30=140$.

Answer with units: 140 kPa.

Physical interpretation: At (70 °C,140 kPa), liquid and vapor can coexist. This interpolation is justified only by the stated straight-segment model.

Common confusions

A phase diagram is not a graph of temperature versus time. Boundary location does not determine phase fractions. Triple and critical points are different. Below triple-point pressure, the liquid region is absent in this model; this is not the same statement as being below the critical pressure.

Real-life connection

Lower atmospheric pressure changes the boiling temperature, which can affect cooking. Pressure-temperature reasoning also explains why temperature alone may not identify a substance’s phase.

Recap

Locate a coordinate, identify its region or boundary, then follow a stated path. Use the substance-specific diagram and keep phase equilibrium separate from heating rates and heat amounts.

Review questions

1. How does a horizontal path differ physically from a vertical path?

2. What information does a boundary point fail to tell you about phase amounts?

3. Why should a melting-line slope drawn for X not be copied onto a water diagram?

Practice

Open WS12. Complete the chapter examples first, then attempt the worksheet independently.

Mastery checklist

☐ I can identify regions, three boundaries and both special points.

☐ I can describe a path with the correct transition names.

☐ I can interpolate only when allowed and state the limits of the diagram.

U2-C07 - Ideal gas law and units

Unit 2 | All competition levels | public-student | Rule alignment: 5.c.ii, printed B57

Learning objectives

Explain the particle basis of PV=nRT; choose consistent pressure, volume and temperature units; solve for one unknown and test whether the ideal model is appropriate.

Prerequisites

Unit 1 Kelvin conversion and algebra; U2-C01 particles and U2-C06 phase conditions.

From collisions to a useful equation

A gas exerts pressure because its particles collide with the walls of the container. Add more particles to the same volume at the same temperature and collisions generally become more frequent, raising the pressure. Heat a fixed gas sample at fixed volume and its particles transfer more momentum in collisions on average, also raising pressure. Give that fixed sample more volume at the same temperature and the pressure falls.

The ideal gas law combines these relationships: PV=nRT. Pressure P and volume V describe the container-scale state. Temperature T describes the thermal state and must be absolute, in Kelvin. The amount n tells us how many particles are present in moles. R is the gas constant connecting the chosen units. The equation is a model of a gas, not an equation for liquid water or ice.

An ideal gas treats particles as having negligible volume relative to the container and negligible interactions except ideal elastic collisions. Real gases often behave approximately ideally when relatively dilute and well away from condensation. High pressure, low temperature near a phase boundary, and strong intermolecular effects can make deviations important. The model is not equally accurate under all conditions, and the equation alone cannot prove a gas will remain a gas during a large change.

A mole counts particles, not grams

A mole is a counting unit, like a dozen but enormously larger. One mole contains exactly 6.02214076×10^{23} specified entities. In ordinary problems, the problem may simply give n. Equal numbers of moles of different gases contain equal numbers of the specified gas particles, but their masses need not match.

If mass and molar mass are provided, n=m/M. Here M is mass per mole, such as g/mol. Thus 4.00 g of a gas with M=4.00 g/mol is 1.00 mol; 4.00 g of a gas with M=40.0 g/mol is 0.100 mol. Do not insert grams directly where PV=nRT asks for moles. No periodic-table memorization is needed when the problem supplies M.

Select one internally consistent unit set

Use R=8.314 kPa·L/(mol·K) with P in kPa and V in L. The same numerical R=8.314 J/(mol·K) works with P in Pa and V in m^3 because 1 Pa·m^3=1 J. It also works because 1 kPa·L=1 J. Another common version is R=0.08206 L·atm/(mol·K), paired with atmospheres and liters. Never combine a constant from one row with pressure/volume units from another without conversion.

Useful conversions are 1 kPa=1000 Pa, 1 L=0.001 m^3 and 1 atm=101.325 kPa. Check that both sides of the equation have matching units. With kPa and L, nRT has units mol×kPa·L/(mol·K)×K=kPa·L, matching PV. This dimensional check often catches a factor-of-1000 mistake.

Use absolute pressure, measured relative to vacuum. A gauge pressure is relative to surrounding atmospheric pressure. If a problem supplies gauge pressure, add the local atmospheric pressure to obtain absolute pressure. For example, a 20 kPa gauge reading in 100 kPa surroundings means 120 kPa absolute. A sealed gas container at room pressure does not have zero absolute pressure simply because a gauge reads zero.

Kelvin is essential

The gas law relates temperature to an absolute zero. Convert with $T_K=T_C+273.15$. A gas at 20 °C is at 293.15 K. Doubling from 20 °C to 40 °C is not doubling absolute temperature, so it will not double pressure at fixed n and V. A change of 20 °C is a change of 20 K, but the ratio uses the absolute readings, not their difference.

To solve for P, divide by V: P=nRT/V. For V, divide by P: V=nRT/P. For n, divide by RT: n=PV/(RT). For T, divide by nR: T=PV/(nR), giving Kelvin if you used consistent units. A final conversion to Celsius may follow if requested. Show the rearrangement before substitution so the unknown's role is clear.

Compare two states carefully

For a fixed amount of ideal gas, PV/T is constant. Therefore $P_1V_1/T_1=P_2V_2/T_2$. If V is fixed, pressure is proportional to Kelvin temperature. If T is fixed, pressure and volume change inversely. If P is fixed, volume is proportional to Kelvin temperature. Each shortcut needs the stated fixed quantities. A leaking balloon is not automatically a fixed-n system.

For Science Olympiad reasoning, first name what stays fixed, then predict the direction, then calculate. If a fixed-temperature gas is compressed to half its volume, the pressure should double in the model; a result that halves it signals an algebra error. If a calculated temperature or pressure is extreme, reconsider the ideal approximation and phase conditions rather than treating a calculator output as proof of physical feasibility.

Definitions and units

P: absolute gas pressure, kPa or Pa or atm. V: gas volume, L or m^3. n: amount of gas, mol. T: absolute temperature, K. R: gas constant with units matching P and V. M: molar mass, g/mol. Gauge pressure: pressure relative to local surroundings, not vacuum.

Equations and their meaning

PV=nRT; P=nRT/V; V=nRT/P; n=PV/(RT); T=PV/(nR). n=m/M when mass and molar mass are supplied. For fixed n, P_1V_1/T_1=P_2V_2/T_2. Use absolute pressure and Kelvin readings. In this unit, R=8.314 kPa·L/(mol·K) unless otherwise stated.

Worked example: Find pressure from amount, volume and temperature

Given: n=0.200 mol; V=5.00 L; T=300 K; R=8.314 kPa·L/(mol·K); ideal gas.

Find: Absolute pressure.

Equation: P=nRT/V.

Substitution: P=(0.200×8.314×300)/5.00 kPa.

Arithmetic: Numerator=498.84 kPa·L; 498.84/5.00=99.768.

Answer with units: P≈99.8 kPa.

Physical interpretation: The result is near ordinary atmospheric pressure; the units are consistent.

Worked example: Find volume after converting Celsius

Given: n=0.100 mol; T=27.0 °C; P=100 kPa absolute; R=8.314 kPa·L/(mol·K).

Find: Gas volume.

Equation: T_K=T_C+273.15; V=nRT/P.

Substitution: T=300.15 K; V=(0.100×8.314×300.15)/100 L.

Arithmetic: 8.314×300.15=2495.4471; V=2.4954471.

Answer with units: V≈2.50 L to three significant figures.

Physical interpretation: Using 27 instead of 300.15 would greatly underestimate the volume.

Worked example: A fixed-volume temperature comparison

Given: A sealed rigid vessel has ideal gas at 90.0 kPa absolute and 300 K; it warms to 330 K with unchanged n.

Find: Final absolute pressure.

Equation: P_2=P_1T_2/T_1 at fixed n,V.

Substitution: P_2=90.0×330/300 kPa.

Arithmetic: 330/300=1.10; 90.0×1.10=99.0.

Answer with units: P_2=99.0 kPa.

Physical interpretation: A 10% Kelvin-temperature increase gives a 10% pressure increase in the model.

Common confusions

Use Kelvin readings, absolute pressure and moles. R changes its numerical form with units; it is not unitless. Fixed-mass comparisons fail if gas escapes. PV=nRT does not describe condensed phases or automatically account for vaporization.

Real-life connection

Temperature changes affect gas pressure in sealed equipment and gas volume in flexible containers. These are written reasoning contexts here, not instructions to heat sealed vessels. Competition water itself is modeled as liquid, while the surrounding air is a gas.

Recap

The ideal gas law combines particle amount, temperature, volume and absolute pressure. Unit consistency and model conditions are as important as arithmetic.

Review questions

1. Why does 20 °C to 40 °C not double the absolute temperature?
2. Which R is consistent with kPa and L?
3. When does the fixed-n combined gas law fail for a real container?

Practice

Open WS13. Complete the chapter examples first, then attempt the worksheet independently.

Mastery checklist

☐ I can choose a consistent R, convert to Kelvin and use absolute pressure.

☐ I can solve PV=nRT for any one unknown.

☐ I can state both the fixed quantities and the ideal-model limitations.

Unit 2 formula and constant packet

public-student | All competition levels | Adds F09–F18; F01–F08 remain unchanged.

Values used when specified: c_water=4.18 J/(g·°C); c_ice=2.10 J/(g·°C); L_f,water=334 J/g near 0 °C and 1 atm; L_v,water=2260 J/g near 100 °C and 1 atm. These rounded teaching values depend on phase and conditions. They are not exact universal material constants.

F09 - Sensible heat

$Q=mc(T_f-T_i)$

Variables: Q: J; m: g; c: J/(g·°C); T_f,T_i: °C

Use: Single phase; approximately constant c; stated heat model. Positive Q enters sample.

Mini example: 20 g×4.18 J/(g·°C)×5 °C=418 J.

F10 - Temperature from heat

$\Delta T=Q/(mc)$; $T_f=T_i+\Delta T$

Variables: ΔT: °C or K change; Q: J; m: g; c: J/(g·°C)

Use: Find a rise before adding it to the starting reading; verify no phase boundary is crossed.

Mini example: 836 J/(40 g×4.18 J/(g·°C))=5 °C; from 19 °C gives 24 °C.

F11 - Transition heat and partial transition

$|Q|=mL$; $m_{changed}=|Q|/L$

Variables: Q: J; m: g; L: J/g

Use: Use the correct transition and conditions. Heating transitions are positive for sample; reverse transitions negative.

Mini example: 2 g×334 J/g=668 J to melt ice at its melting point.

F12 - Stage-by-stage energy

$Q_{total}=Q_1+Q_2+\ldots$

Variables: Q_i: J per stage

Use: Separate warming/cooling from each phase change. Use a different c for each phase when needed.

Mini example: Ice warming 42 J + melting 668 J + liquid warming 83.6 J =793.6 J.

F13 - Energy at constant net input

$Q = P_{net}\Delta t$

Variables: Q: J; P_{net}: W=J/s; Δt: s

Use: P_{net} is the stated transfer into the sample, not automatically a lamp's electrical rating.

Mini example: 50 W×120 s=6000 J.

F14 - Average graph slope

$\text{rate} = (T_2 - T_1)/(t_2 - t_1)$

Variables: T: °C; t: min; rate: °C/min

Use: Read both axes; a cooling rate is negative under final-minus-initial convention.

Mini example: (40−20) °C/(6−2) min=5 °C/min.

F15 - Ideal gas law

$PV = nRT$

Variables: P: absolute kPa; V: L; n: mol; T: K; R: 8.314 kPa·L/(mol·K)

Use: Ideal-gas approximation; Kelvin and absolute pressure. Alternative matched sets use Pa/m³ or atm/L with matching R.

Mini example: 0.100 mol×8.314×300 K / 2.00 L=124.71 kPa≈125 kPa.

F16 - Mass to moles

$n = m/M$

Variables: n: mol; m: g; M: g/mol

Use: M is supplied molar mass; do not substitute grams for moles.

Mini example: 6.00 g/(12.0 g/mol)=0.500 mol.

F17 - Two ideal-gas states

$P_1V_1/T_1 = P_2V_2/T_2$

Variables: P_1, P_2: same absolute-pressure unit; V_1, V_2: same volume unit; T_1, T_2: K

Use: Requires constant n and ideal behavior; simplify only quantities explicitly fixed.

Mini example: At fixed n,V: 100 kPa×330 K/300 K=110 kPa.

F18 - Gas-law unit and pressure conversions

1 kPa=1000 Pa; 1 L=0.001 m^3; 1 atm=101.325 kPa; $P_{abs}=P_{gauge}+P_{atm}$

Variables: P_{abs},P_{gauge},P_{atm}: same pressure unit

Use: Pressure in $PV=nRT$ is relative to vacuum. 1 kPa·L=1 Pa·m^3=1 J.

Mini example: A 15 kPa gauge reading with local atmosphere 99 kPa gives 114 kPa absolute.

Unit 2 glossary

public-student | All levels | Complements the unchanged Unit 1 glossary.

Absolute pressure: Pressure measured relative to a vacuum; the pressure required by PV=nRT.

Boiling: Vapor bubbles form and grow within a liquid when vapor pressure reaches surrounding pressure under the stated conditions.

Condensation: Gas-to-liquid transition; releases heat from the condensing sample under the usual thermal transition model.

Critical point: Endpoint of the liquid-gas coexistence curve, beyond which the ordinary liquid-gas distinction disappears.

Deposition: Direct gas-to-solid transition.

Endothermic phase change: Transition that absorbs heat into the selected sample, such as melting.

Evaporation: Liquid-to-gas transfer at the surface; can occur below boiling temperature.

Exothermic phase change: Transition that releases heat from the selected sample, such as freezing.

Freezing: Liquid-to-solid transition.

Fusion: Another name for melting in latent-heat terminology; not nuclear fusion here.

Gauge pressure: Pressure measured relative to the surrounding atmosphere.

Ideal gas: Model with negligible particle volume and negligible interactions except ideal collisions.

Latent heat: Transition heat per mass at specified conditions, usually J/g or J/kg.

Melting: Solid-to-liquid transition.

Molar mass: Mass per mole of specified entities, often g/mol.

Mole: Amount containing exactly $6.02214076 \times 10^{23}$ specified entities.

Net heating power: Effective rate of heat input into the sample; not necessarily the source power rating.

Phase boundary: Pressure-temperature conditions where adjacent phases can coexist in equilibrium.

Phase diagram: Map of stable equilibrium phases as a function of conditions, usually pressure and temperature.

Plateau: Approximately horizontal temperature segment; an ideal phase transition can produce one at fixed pressure.

Sensible heat: Heat associated with a temperature change in the specified model.

Specific heat: Heat per mass per temperature change under specified conditions.

Sublimation: Direct solid-to-gas transition.

Supercritical fluid: Fluid above the critical temperature and pressure without a distinct ordinary liquid-gas boundary.

Supercooling: A liquid remaining liquid temporarily below its equilibrium freezing temperature.

Triple point: Particular temperature and pressure permitting solid, liquid and gas coexistence in equilibrium.

Vaporization: Liquid-to-gas transition; includes evaporation and boiling.

Vapor pressure: Equilibrium pressure of a substance’s vapor above its condensed phase at a given temperature.

Unit 2 review sheet

public-student | All competition levels | WS07–WS13

Choose the model before substituting numbers

Evidence in the question	Model or reasoning	Check first
Single phase changes temperature	$Q=mc\Delta T$	Correct phase, c, mass unit and final-minus-initial
Pure sample changes phase at transition conditions	$Q=\pm mL$	Correct transition and amount that actually changes
Several stages	One energy term per stage	Finish a stage before spending remaining energy
Time-temperature data with constant net power	Slope=$\Delta T/\Delta t$; $Q=P_{net}\Delta t$	Axis units; seconds for W×time
Pressure-temperature diagram	Regions, boundaries and paths	Substance-specific diagram; equilibrium model
Gas with P,V,n,T	$PV=nRT$	Kelvin, absolute pressure, matching R and ideal approximation

Seven conceptual checkpoints

1. Solid particles vibrate, liquid particles rearrange among close neighbors, and gas particles spread through available space. A state change need not change chemical identity or mass.

2. Melting, vaporization and sublimation absorb heat in the usual thermal transition model. Freezing, condensation and deposition release it. Evaporation can happen below boiling.

3. A plateau can receive or release energy while temperature stays constant. Its duration is not itself a material constant.

4. ΔT is a change, not a final reading. Specific heat and conductivity answer different physical questions.

5. Compare the available heat with transition requirements before deciding the final phase. Ice at 0 °C need not be melted.

6. A phase boundary permits coexistence but does not specify the mass fractions. Triple and critical points are different.

7. Gas-law ratios need Kelvin. Moles count entities; grams measure mass. A gauge reading is not absolute pressure.

Competition-style explanation routine

State the model, identify the relevant evidence, show the equation or graph path, and check the physical result. A correct number with the wrong model is fragile knowledge. For a drawing, label axes or system boundaries and say what the picture leaves out.

Remediation map

Particle spacing/identity → WS07. Transition names and direction → WS08. Slopes/plateaus → WS09. Q=mcΔT and signs → WS10. Partial transitions/stages → WS11. Diagram coordinates/coexistence → WS12. Kelvin/R/moles/pressure → WS13.

Ready to advance?

Aim for at least 16/20 per worksheet and explain corrected errors. This is a local teaching target, not official scoring. Be able to justify one graph answer and one energy budget without memorizing a solution. QZ02 and T01 remain scheduled for the later assessment phase.